

Spending Across the 2010's

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Background:

Understanding spending behavior is important for businesses to remain competitive within the economic market. Analyzing when customers spend the most, their preferred transaction methods, and how spending varies across age groups can provide insights into financial trends and decision-making patterns. Following the financial recession, the 2010s provided insight into a period of economic recovery, during which customer spending began recovering. At the same time, advancements in technology led to a significant increase in online transactions and digital payment methods. These changes reshaped how customers spend and manage their finances. By studying transaction data across this period, this project aims to better understand seasonal spending patterns, long-term trends, and demographic differences in financial behavior among age groups.

Outline:

Our course project uses the Financial Transactions Dataset: Analytics (2009 - 2019) from Kaggle: <https://www.kaggle.com/datasets/computingvictor/transactions-fraud-datasets>

The goal of our project was to extract data to visualize and understand changes in consumer spending behaviours over the course of a year and during periods of economic instability. We chose this dataset because it includes anonymized financial transaction records over a 10-year period, including details about whether a transaction was fraudulent. The dataset components consist of transaction data, card information, merchant category codes, fraud labels, and user data.

The key features we focused on were:

- Transaction amount
- Transaction date
- Transaction method (swipe, chip, online)
- User age
- Credit limit

This dataset is well suited for analyzing financial trends because it covers the economic recovery following the recession in 2009, along with the greatest period of growth for online transactions during the 2010s.

Major Questions:

For this project, we extracted data to analyze and answer the following questions:

- What times of year are busiest for commerce?
- How does spending change in the years following a recession (2010-2019)?
- How does the volume of online spending change as the internet matures?
- How does spending change across age groups on average, and who spends the highest percentage of their available credit?

Tools:

For our project, we used PostgreSQL, Python, Tableau, and Docker.

- PostgreSQL - Used to store and manage our cleaned data, allowing for efficient queries across the dataset.
- Python - Used to clean our data and perform data analysis. Python has strong libraries for working with financial and tabular data, making it ideal for our project.
- Tableau - Used for our final report and visualization. Paired with our use of Python and PostgreSQL, Tableau made it easy to visualize and analyze our data further after filtering and cleaning.
- Docker - Used to create and run a reproducible PostgreSQL environment for the project.

Data Pipeline:

- Data acquisition/loading: Downloaded the dataset from Kaggle and prepared the raw files for use in the project.
- ETL / cleaning: Removed formatting issues, standardized values, converted data types, and cleaned fields before loading the data into PostgreSQL. This included handling missing values and invalid placeholders where necessary.
- Storage of cleaned data: Stored cleaned data in PostgreSQL.
- Exploratory Data Analysis (EDA): Used summary statistics, trend-based views, and grouped comparisons to examine transaction patterns over time, transaction methods, and age groups.
- Reporting: Created visuals and dashboards in Tableau to present the final results.

Ethics and Considerations:

Since our project uses anonymized transaction data, there are no direct privacy concerns for vendors or customers in any given transaction. Some ethical concern could be given to the usage of metrics extracted from this data, as exploiting consumer spending habits could be considered unethical. This project is purely analytical and offers no advice for exploiting consumer practices.

The source dataset is licensed under the Apache 2.0 conditions, meaning we are free to use it as long as the source is properly credited and our use remains consistent with the license terms.

Results and Discussion:

What times of year are busiest for commerce?

To find the busiest times of year for commerce, we examined monthly spending patterns across each year in the dataset, shown in Figure 1. The results show a fairly consistent monthly pattern each year, with February consistently appearing as the weakest month for spending. Spending increases and peaks in March, May, July, October, and December, with December appearing to be one of the strongest months overall. This suggests that commerce follows recurring monthly trends rather than changing randomly throughout the year.

How does spending change in the years following a recession (2010-2019)?

To see how spending changed in the years following the recession, we examined the total spending volume over time from 2010 to 2019, shown in Figure 1. The results suggest that spending generally increased in the years following the recession, particularly from 2010-2014. After 2015, spending appeared to level off at a higher overall level. This supports the idea that consumer spending recovered over time following the recession and then became more stable later in the 2010s.

How does the volume of online spending change as the internet matures?

To explore how online spending changed as the internet matured, we examined online transaction volume across the decade, shown in Figure 1. The results show a clear upward trend in online transaction amounts from the early 2010s to around 2015. After 2015, the online transaction amount begins to level off, remaining higher than it was at the beginning of the time period. This suggests that online transactions became a more important part of overall consumer spending as digital commerce continued to grow.

How does spending change across age groups on average, and who spends the highest percentage of their available credit?

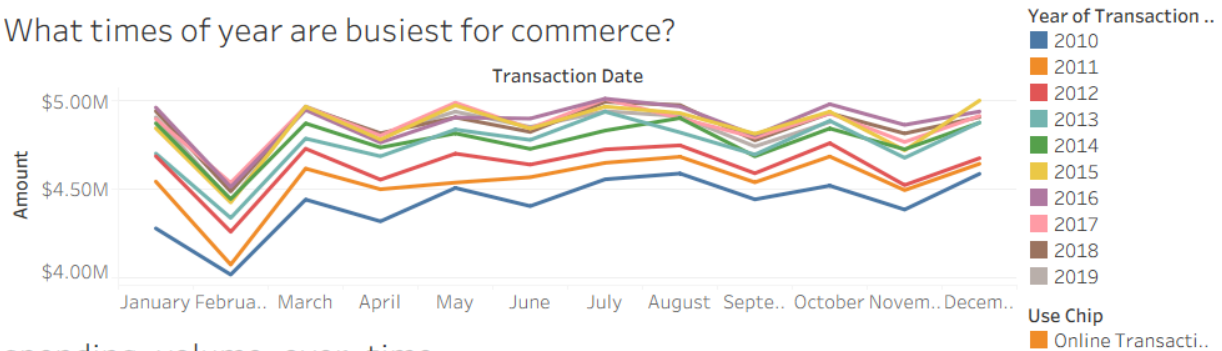
To find how spending changed across age groups and which group used the highest percentage of available credit, we compared both spending per client and credit usage by age group, shown in Figure 2. The results suggest that older age groups, especially ages 45 to 54 and 55+, generally had the highest spending per client. In contrast, the 18 to 24 age group showed the highest credit usage percentage, often by a large margin compared to the other age groups. This suggests that while older individuals may spend more overall, younger people tend to use more of their available credit.

Exploratory Data Analysis.

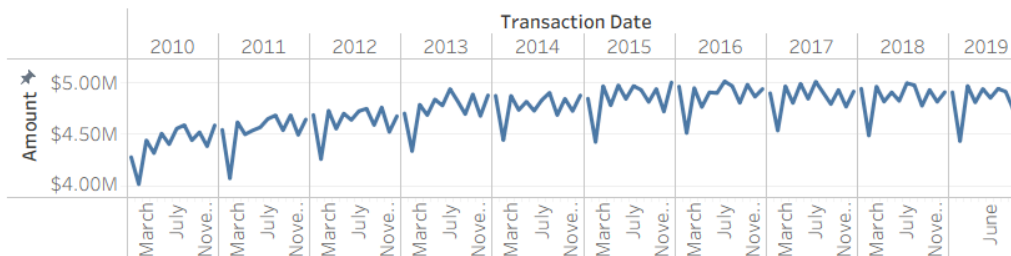
Additionally, Figure 1 provides an exploratory summary of the dataset through a heatmap of transaction counts by month and year, yearly household spending distributions, credit availability over time, total spending per client, and a transaction location map. This exploratory view helped

provide context for the broader trends examined in the later figures and supported the decision to focus on seasonality, long-term spending growth, online transactions, and age-group differences.

What times of year are busiest for commerce?



spending_volume_over_time



online_transactions_by_year_month

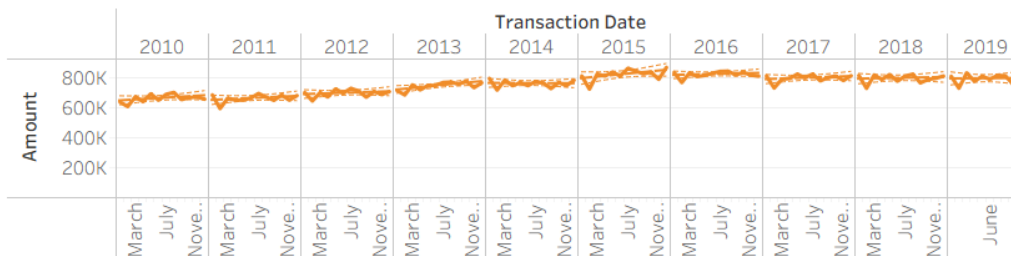
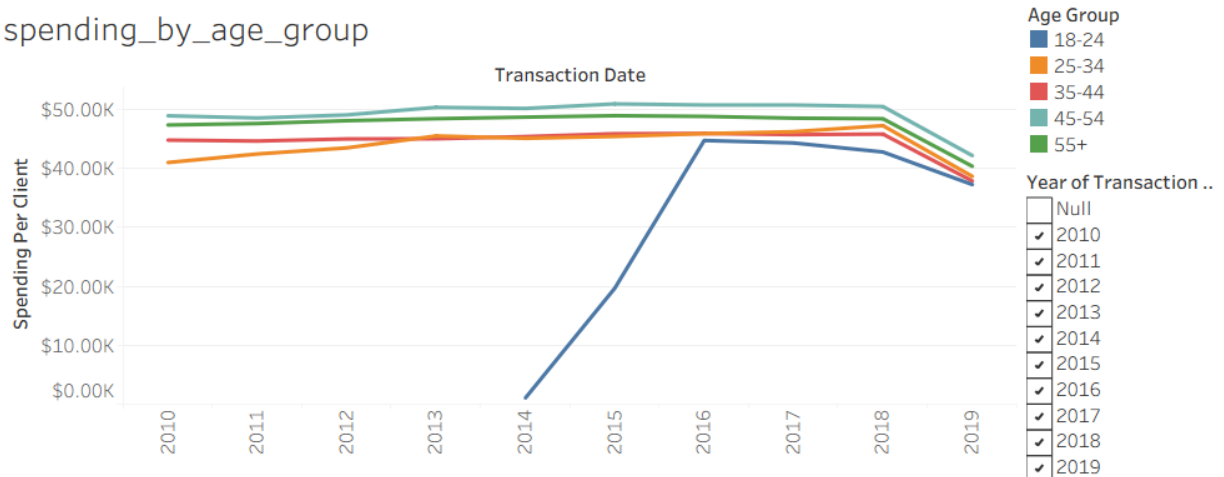


Figure 1. Monthly and yearly spending trends, including seasonal transaction patterns, total spending volume, and online transaction growth from 2010 to 2019.

spending_by_age_group



credit_usage_per_age_group

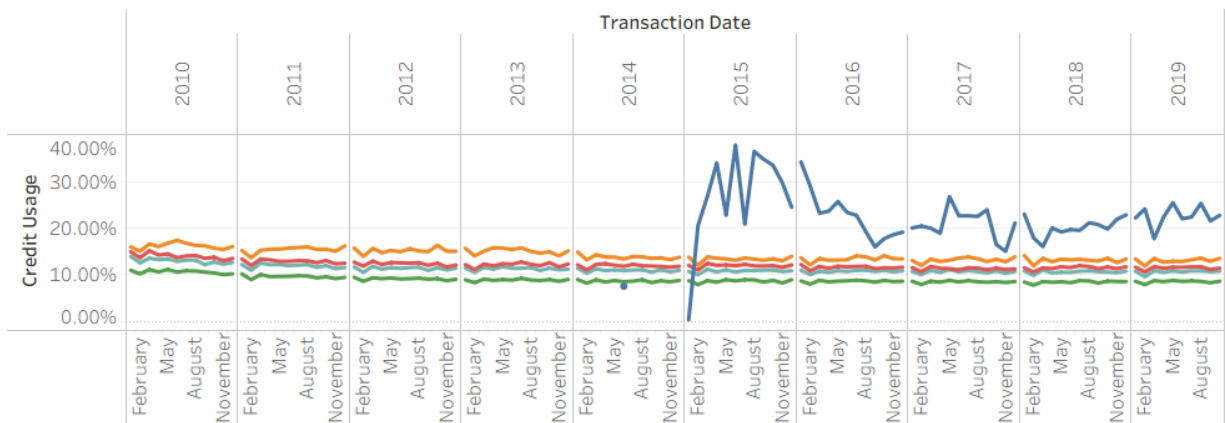
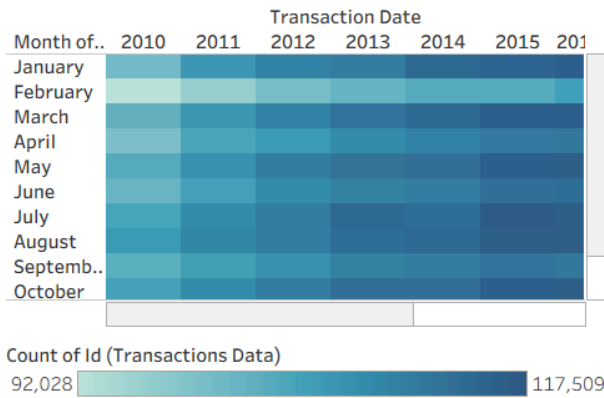
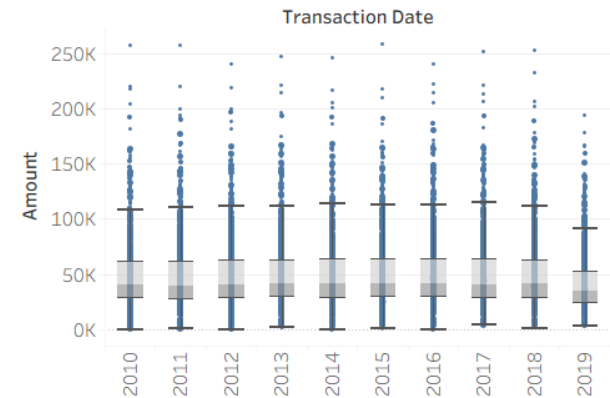


Figure 2. Comparison of spending per client and credit usage across age groups over time.

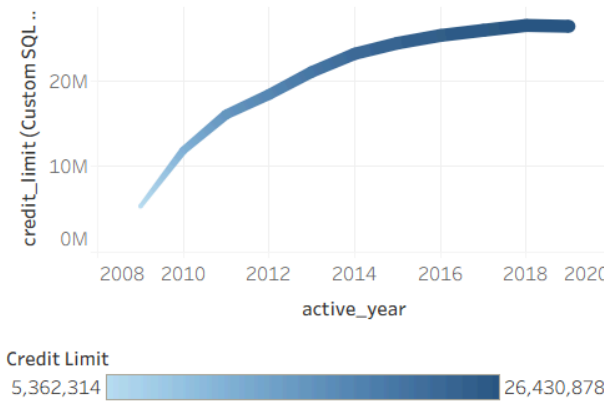
transactions_ranges_per_year



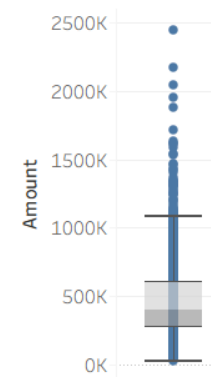
household_spending_summary



credit_available_by_year



total_spend_per_client



spending_map



Figure 3. Exploratory data analysis of transaction patterns, household spending distributions, available credit over time, spending per client, and transaction geography.